LAB 09 - PROMPTS

Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab09-performance-metrics/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Read the metric formulas (K3)

Prompt - copy into your AI coding assistant
Explain each function in labs_fullcode/common/metrics.py with its formula and one sentence on what a portfolio manager learns from it; say which metrics need trade-level returns and which need a benchmark. Do not change the file.

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Check: You can explain the difference between Sharpe (all volatility) and Sortino (downside only).

Step 2: Build the evaluation stage (A3, K3)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/evaluate.py. evaluate_strategy(bars, name, costs, start, bench_ret, **params) backtests pipeline.signals.target_position(name, bars, **params) with common.backtest.backtest (next-bar open fills, costs = pipeline.config.cost_model(cfg)), slices returns and position from `start` and returns common.metrics.summary(...) with trade net returns and the benchmark. run(symbol='SPY'): start = first date the 200-day SMA exists (so every signal is warm), benchmark = buy_and_hold sliced from start; evaluate every signal in SIGNALS; print a table (rows = strategies + buy & hold, columns CAGR, Ann. volatility, Sharpe, Sortino, Max drawdown, Calmar, Trades, Win rate, Profit factor, Exposure, Turnover, Alpha, Beta) ranked by Sharpe and formatted with common.metrics.format_table; list which strategies beat buy and hold on Sharpe and how many have a smaller drawdown; print 'Sharpe by hand' for trend_filter (mean / stdev x sqrt(252)), the worst day and SPY's drawdown trough; save _outputs/Lab09_metrics.csv and chart Lab09_metrics.png (top 4 by Sharpe plus buy & hold, equity and drawdown). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline evaluate --help shows --symbol.

Step 3: Run the evaluation (A3)

Do this (no prompt needed)
Run: python -m pipeline evaluate --symbol SPY (same as: python3 Lab09-performance-metrics/evaluate_stage.py).

Check: Window 2018-10-16 to 2025-12-31, Alpaca-style costs. ema_cross: CAGR 11.09%, vol 11.87%, Sharpe 0.95, max drawdown -14.19%, Calmar 0.78, 26 trades, win rate 50.00%, profit factor 3.54, beta 0.35. SPY buy & hold: 15.28%, 19.93%, 0.81, -33.72%, 0.45. trend_filter: 9.70%, 12.42%, 0.81, -19.36%, 24 trades, win 41.67%, profit factor 3.79, exposure 78.53%, beta 0.38. Only ema_cross beats buy & hold on Sharpe; 7 of 8 have a smaller drawdown.

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Step 4: Check one metric by hand (K3)

Do this (no prompt needed)
Read the 'Sharpe by hand' line and confirm it equals the table.

Check: mean 0.00040 / stdev 0.00782 x sqrt(252) = 0.81, the same as trend_filter's Sharpe in the table.

Step 5: Interpret trade and market metrics (A3, K3)

Prompt - copy into your AI coding assistant
Using labs_fullcode/_outputs/Lab09_metrics.csv, write a 120-word note for Lion City Capital's investment committee comparing ema_cross, trend_filter and SPY buy & hold on return, risk and risk-adjusted return. Quote the numbers, explain win rate versus profit factor, and do not recommend live trading.

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Check: Few large winners pay for many small losers; the strategy is out of the market in many stress periods, so its beta is far below its exposure.

Step 6: Locate the worst periods (K3)

Do this (no prompt needed)
Read the worst-day and drawdown-trough line and find those dates on _outputs/Lab09_metrics.png.

Check: Worst trend_filter day 2020-06-11 (-5.76%); worst SPY day 2020-03-16 (-10.94%); SPY drawdown trough 2020-03-23.

Step 7: Add a metric to the stage (K3)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/evaluate.py add an 'Ulcer index' column = sqrt(mean(drawdown% squared)) computed from common.metrics.drawdown_series of each strategy's returns, without changing common/metrics.py. Show it in the printed table and the CSV, and explain in one sentence what it adds to max drawdown. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: The new column shows a much lower Ulcer index for ema_cross than for SPY buy & hold.

Run it

Step	Command / file
Run Lab 09	python3 Lab09-performance-metrics/evaluate_stage.py (same as: python -m pipeline evaluate)
The complete code	pipeline/evaluate.py (reference copy: Lab09-performance-metrics/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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